

Cost Sharing in Multiple Use Forestry: A Game Theoretic Approach

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ABSTRACT. Game theory is applied to the problem of joint cost allocation in forest management. The theory is used to define bounds (limits) to cost allocations. These traditional bounds are further defined in the context of potentially relevant benefit information to arrive at "benefit-included" bounds. These benefit-included bounds are consistent with both game theory and the early water resources principles that were based on cost separability. The traditional bounds and the benefit-included bounds are applied to a multipurpose forestry case study. The application is facilitated using the "nucleolus" cost allocation technique. The nucleolus provides a unique allocation that does not breach the bounds. The previously obscure calculation procedure of the nucleolus is clarified by example. *FOR. SCI.* 33(1):81-88.

ADDITIONAL KEY WORDS. Nucleolus, cost allocation, joint production, core theory.

ADMINISTRATORS OR MANAGERS may be required to allocate the costs of multiple purpose management to its different purposes. This often poses a dilemma because many costs of multiple-purpose forestry are properly considered joint.¹ The economically arbitrary nature of joint cost allocation is well established in the natural resources and forestry literature. For example, Herfindahl and Kneese (1974) state "The history of cost allocation is characterized by a long, and unsuccessful, search for a rigorous or theoretically sound method." In the forestry literature, this notion was affirmed by Bowes and Krutilla (1979) in their statement that "the allocation of joint costs is in general arbitrary," by Duerr (1960) and by Hof et al. (1985).

This article applies game theory to joint cost allocation problems in forest management. The article extends traditional gaming concepts to include benefit considerations. Specifically, traditional bounds (limits) on joint cost allocations are discussed and then modified to include benefit information. Although a theoretic framework for cost allocation bounds is discussed and specific allocations are generated, it is not suggested that allocation of joint costs is nonarbitrary.

The framework is applied to a simplified case study of a multipurpose forest road in northern Colorado. Specific allocations are calculated using an allocation technique known as the "nucleolus." The nucleolus was chosen because it, and modifications of it, provide the only available means of guaranteeing unique

¹ Joint costs, as used here, refer to the costs of multipurpose production that are not separable.

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results within the given bounds (Schotter and Schwodiauer 1980). Two contributions are made in applying the nucleolus. First, the procedure for calculating the nucleolus is clarified and simplified—previously published applications involve obscure programming procedures. Second, the nucleolus is modified to include benefit information.

Game theory is used here because it has three main advantages relative to approaches from the early water resources literature [for example, the Federal Inter-Agency River Basin Committee (FIRBC 1958)]. First, game theory defines a set of allocation bounds that specifically account for groups of purposes in addition to the individual purposes accounted for in the early literature. Second, game theory has proved to be a convenient vehicle for expressing and generating specific cost allocations within this more complete set of bounds (Young et al. 1982). Third, the allocation bounds of game theory are better related to concepts of economic equilibria such as Pareto efficiency and the Edgeworth contract curve (Atkinson and Stiglitz 1980).

CORE THEORY AND COST ALLOCATIONS

THE TRADITIONAL CORE FRAMEWORK

The notion of a core was developed in n -person cooperative game theory (Shapley 1953). “The core . . . is the set of feasible outcomes that cannot be improved upon by any coalition of players” (Shapley 1971). This embodies the familiar concept of Pareto efficiency, but accounts for both individuals and groups of individuals (Hamlen, Hamlen, and Tschirhart 1980, Atkinson and Stiglitz 1980). Core solutions include all sets of cost allocations that do not imply disincentives to player cooperation. Elements of the core would be applied to multipurpose projects by defining a set of N project purposes instead of a set of game players.

In multipurpose projects, the core bounds have been defined by alternative costs (Young et al. 1982), where alternative cost is the least cost of separately producing a given individual or group of project purposes. Allocations that satisfy the core conditions do not exceed any alternative costs. As reported by Young et al. (1982) and shown by Rideout and Wagner (1986), the core bounds, defined by alternative costs, can equivalently be defined by the set of separable (nonjoint) costs. This equivalence is used below for descriptive purposes. Alternative costs denote upper allocation bounds and separable costs denote lower allocation bounds.

Separable cost, for individual and groups of purposes, is defined as the cost increment:

$$SC(T) = C(N) - C(S) \quad (1)$$

and

$$\begin{aligned} S \cup T &= N \\ S \cap T &= [] \end{aligned}$$

In (1), $SC(T)$ denotes the separable cost of product set T . N denotes the set of products in a joint project, and S is the set of products in N but not in T . Sets S and T are mathematically complementary. C is the least cost means of producing a specified product set. For example, $C(S)$ is the least cost means of producing product set S . As stated by the FIRBC (1958), “The separable cost for each project purpose is the difference between the cost of the multiple-purpose project and the cost of the project with the purpose omitted.” Equation (1) generalizes this definition to include the separable costs of purpose groups.

The lower and upper cost allocation bounds are given by:

$$SC(i) = C(N) - C(N - i) \leq A(i) \leq C(i), \quad \forall i \in N \quad (2)$$

$$SC(T) = C(N) - C(S) \leq A(1) + \dots + A(t) \leq C(T), \quad \forall T \subset N \quad (3)$$

$$A(1) + A(2) + \dots + A(n) = C(N) \quad (4)$$

In (2) there are n elements in the set N and in (3) there are t elements in the set T . Equation (2) states the individual upper and lower bounds—that an allocation to the i th purpose, $A(i)$, must be greater than or equal to that purpose's separable cost and less than or equal to its alternative cost. The term $C(N - i)$ denotes project cost with the i th purpose removed. Equation (3) states the group upper and lower bounds—that the allocation to any group of purposes must not exceed the group's alternative cost, and must equal or exceed that group's separable cost. Equation (4) states the group bound [implicit in (3)]—that the sum of allocations must equal total project cost. There is an upper and lower bound on each individual purpose and on each group of purposes. For core-consistent allocations, no bounds can be breached. And, again, the core can be defined with either the set of upper bounds (alternative costs) or the set of lower bounds (separable costs).

THE ROLE OF BENEFIT INFORMATION (BENEFIT-INCLUDED CORE)

In establishing the criterion for multipurpose cost allocations, the FIRBC (1958) states "On the assumption that the principles for project formulation recommended herein have been applied, equitable distribution may be obtained by preventing costs allocated to any purpose from exceeding corresponding benefits; (and) by requiring each purpose to carry at least its separable cost . . ." The "traditional" core discussed above ensures separable cost coverage but does not limit allocations to corresponding benefits.

The fundamental principle that allocations should not imply a potential disincentive to joint production (when joint production is efficient) is extended here to apply to benefits. By the logic expressed in FIRBC (1958), if purpose "accounts" are charged more than their benefit, a disincentive to efficient/joint production may be introduced. Where such disincentives are relevant, allocations should not exceed respective benefits. This defines a benefit-included core. Simply stated, allocations consistent with the benefit-included core do not exceed the lesser of benefit and alternative cost [$\min(B, AC)$] for individual and groups of purposes.

If some benefits are less than respective alternative costs, the upper allocation bounds defined by the [$\min(B, AC)$] no longer exactly coincide with the lower allocation bounds defined by the set of separable costs. However, allocations consistent with the benefit-included core cover each separable cost. This makes the benefit-included core consistent with both game-theoretic logic and the cost allocation criterion from the early water resources work (FIRBC 1958). For example, consider a four-purpose project with a total cost of \$100. If the alternative cost of purpose one is \$40, but the corresponding benefit is \$35, then the upper bound to purpose one is \$35. The separable cost of the group two-three-four is \$60 ($100 - 40$). However, two-three-four will be allocated at least \$65 because the upper bound to purpose one is \$35.

Two important qualifications should be noted. First, it is mathematically possible for no core to exist. It is then impossible to find an allocation that would simultaneously meet each upper bound. In this situation, a cooperative solution is impossible, and the theory developed and discussed above does not apply. Second, while the benefit-included core ensures that each individual and group allocation is less than or equal to its benefit, it is inappropriate and potentially misleading to use the cost allocations to economically justify any purpose or group of purposes. Correct economic evaluation relies on differences between total costs and total benefits, not on cost allocation.

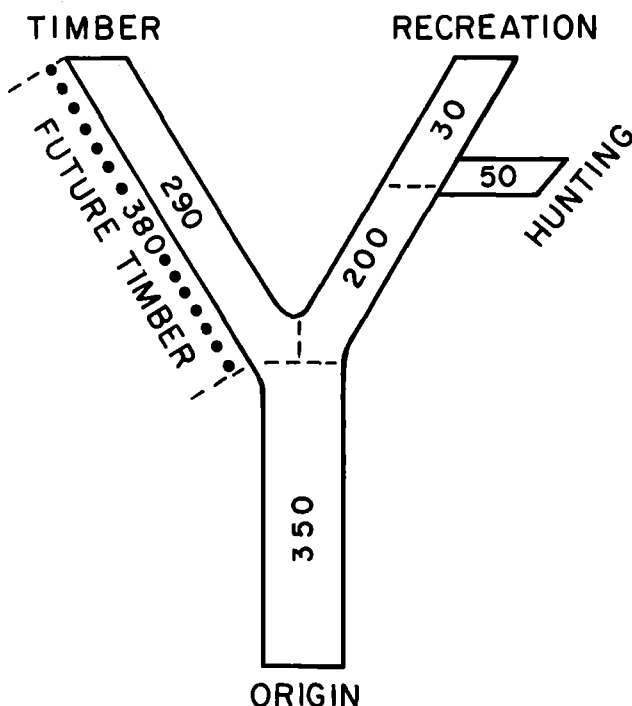


FIGURE 1. Incremental cost structure of a four-purpose forest road (units are thousands of dollars).

COST SHARING APPLIED TO A FOREST ROAD

The nucleolus allocation technique is applied to the cost structure of a forest road under two circumstances. First, benefits are assumed irrelevant regarding the cost allocation bounds, and the nucleolus is directly applied. Second, benefits are assumed relevant and some benefits are assumed less than their alternative costs. In this instance, a benefit-included core is defined, and the nucleolus procedure is modified accordingly. In both instances, *net* project benefits are assumed positive and maximum.

INCREMENTAL COST STRUCTURE OF A FOREST ROAD

The example reflects cost data obtained from a USDA Forest Service timber sale. The forest road is to promote four purposes: timber hauling over the period of the timber sale, recreation traffic, hunting traffic, and future timber hauling on subsequent sales. Project cost estimates are shown in Figure 1 and listed in Table 1. The cost of providing for jointly accessed road segments (other than the jointness between timber and future timber) is assumed independent of the number of purposes included. This simplifying assumption is made so the cost structure is readily visible.

The only cost increment not visible in Figure 1 is the separable cost of the group timber-future timber (\$460,000). This cost can be calculated from Figure 1 by subtracting the alternative cost of the group recreation-hunting (30,000 + 50,000 + 200,000 + 350,000) from the total project cost (\$1,090,000). Likewise, the alternative cost of timber-future timber (\$810,000) listed in Table 1 can be derived from Figure 1 by subtracting the separable cost of the group recreation-hunting (200,000 + 30,000 + 50,000) from total cost (\$1,090,000).

TABLE 1. *Alternative cost estimates for a four-purpose forest road (units are thousands of dollars).*

Product combination	Alternative costs
Single product estimates	
C(T) = Timber	640
C(R) = Recreation	580
C(H) = Hunting	600
C(F) = Future timber	730
Double product estimates	
C(TR) =	870
C(TH) =	890
C(TF) =	810
C(RH) =	630
C(RF) =	960
C(HF) =	980
Triple product estimate	
C(TRH) =	920
C(TRF) =	1,040
C(THF) =	1,060
C(RHF) =	1,010
Project cost estimate	
C(TRHF) =	1,090

The cost estimates in Table 1 reflect least cost production (alternative costs). The cost of providing for timber hauling, with none of the other project outputs, $C(T)$, is \$640,000. Similarly, the cost of providing for just timber and recreation traffic during the sale period is listed as $C(TR)$ or \$870,000. The remaining incremental cost data are similarly interpreted. Establishing a cost category for future production is a form of amortization. In this problem, a minimum of \$170,000 ($1,090,000 - 920,000$) of total project cost is separable to future hauling. More extensive amortization would be possible by subdividing the future into additional time periods.

NUCLEOLUS ALLOCATIONS

While the core boundaries, equations (2) through (4), define an allocation region, the nucleolus is defined by a unique point in the core. For any core, there is a single nucleolus. The nucleolus has been described as:

that cost sharing imputation that minimizes the maximum complaint that any coalition could have against it, where a complaint in this context measures the difference between the sum of fees charged to a coalition and the total cost to that coalition of providing the service under consideration to itself (alternative cost) . . . the nucleolus (has been compared) to a Rawlsian cost-sharing scheme whose purpose is to maximize lexicographically the payoff of the worst off player. (Schotter and Schwodiauer 1980)

The nucleolus is perhaps best defined by example. It is often solved by using a series of linear programs (LPs). Although such a series immediately follows and some intermediate values are produced, only the final set of values are the nucleolus.

The first LP in the series for the forest road problem is denoted as:

Maximize L , subject to:

$$A(T) + A(R) + A(H) + A(F) = 1,090,000 \quad (5)$$

$$A(T) + L \leq 640,000 \quad (6)$$

$$A(R) + L \leq 580,000 \quad (7)$$

$$A(H) + L \leq 600,000 \quad (8)$$

$$A(F) + L \leq 730,000 \quad (9)$$

$$A(T) + A(R) + L \leq 870,000 \quad (10)$$

$$A(T) + A(H) + L \leq 890,000 \quad (11)$$

$$A(T) + A(F) + L \leq 810,000 \quad (12)$$

$$A(R) + A(H) + L \leq 630,000 \quad (13)$$

$$A(R) + A(F) + L \leq 960,000 \quad (14)$$

$$A(H) + A(F) + L \leq 980,000 \quad (15)$$

$$A(T) + A(R) + A(H) + L \leq 920,000 \quad (16)$$

$$A(T) + A(R) + A(F) + L \leq 1,040,000 \quad (17)$$

$$A(T) + A(H) + A(F) + L \leq 1,060,000 \quad (18)$$

$$A(R) + A(H) + A(F) + L \leq 1,010,000 \quad (19)$$

This LP maximizes the smallest deviation (L) from any upper bound as denoted by equations (6) through (19), while exactly meeting (5). The lower bounds (separable costs) are fully accounted for in equations (6) through (19) because of the set complementarity expressed in equations (1) through (4). This LP has multiple optima, but one specific optimal solution (that obtained) is: $L = 175,000$, $A(T) = 290,000$, $A(R) = 230,000$, $A(H) = 225,000$ and $A(F) = 345,000$. An optimal solution to this first LP is known as the “least core” (Young et al. 1982), and it is often not unique. In this case, constraints (5), (12), and (13) are binding.

The second step (LP) is maximization of the next to the smallest deviation from any upper bound (6) through (19) while retaining the smallest deviation (L) in the previous LP. Thus, an LP identical to the one above is solved, except constraints (12) and (13) are replaced with:

$$A(T) + A(F) \leq 635,000 \quad (12a)$$

$$A(R) + A(H) \leq 455,000 \quad (13a)$$

L now represents the next to the smallest deviation from any core condition. In (12a) and (13a), the L is removed and the right hand sides are reduced from those in (12) and (13) by the initial solution value for L . A solution to this second LP, which also has multiple optima is: $L = 187,500$, $A(T) = 277,500$, $A(R) = 217,500$, $A(H) = 237,500$ and $A(F) = 357,500$.

In principle, the nucleolus is arrived at by continuing this process until all of the deviations from the core conditions have been sequentially maximized. In practice, this process need continue only until a multiple optima is not encountered. In this case, the *third* iteration results in a unique optimal solution—the nucleolus. The nucleolus will be arrived at through this procedure, regardless of which multiple optima are obtained above. The nucleolus allocations are displayed in Table 2 relative to the upper and lower bounds.

BENEFIT-INCLUDED NUCLEOLUS

When benefits are relevant and some are less than alternative costs, the nucleolus procedure can be appropriately modified. Benefit estimates of: timber \$800,000, recreation \$825,000, hunting \$125,000, and future timber \$330,000 are assumed. These benefits are assumed independent and additive. The $[\min(B, AC)]$ are listed in Table 3, and comparison with Table 2 shows some substantial reductions in the upper bounds. The bounds were reduced on hunting (from \$600,000 to

TABLE 2. Nucleolus allocations relative to core conditions for a four-purpose forest road example (units are thousands of dollars).

Purpose(s)	Lower bound (Separable costs)	Nucleolus	Upper bound (Alternative costs)
T	80.0	272.5	640.0
R	30.0	217.5	580.0
H	50.0	237.5	600.0
F	170.0	362.5	730.0
TR	110.0	490.0	870.0
TH	130.0	510.0	890.0
TF	460.0	635.0	810.0
RH	280.0	455.0	630.0
RF	200.0	580.0	960.0
HF	220.0	600.0	980.0
TRH	360.0	727.5	920.0
TRF	490.0	852.5	1,040.0
THF	510.0	872.5	1,060.0
RHF	450.0	817.5	1,010.0
TRHF	1,090.0	1,090.0	1,090.0

\$125,000), on future timber (from \$730,000 to \$330,000), and on the group hunting-future timber (from \$980,000 to \$455,000).

The benefit-included nucleolus was obtained using the same procedure as for the nucleolus, but benefits replace alternative costs where benefits are lower. In this example, the right-hand side denoted in (8) was reduced from \$600,000 to \$125,000, and the right-hand sides in (9) and (15) were reduced to their benefit levels.

The $[\min(B, AC)]$, the separable cost set, and the benefit-included nucleolus allocations are listed in Table 3. The values in Table 3 illustrate two important points. First, if the benefit constraints are binding, the smallest deviation between

TABLE 3. The benefit-included core and benefit-included nucleolus allocations when some benefits are less than respective alternative costs (units are thousands of dollars).

Product	Lower bound (Separable costs)	Benefit-included nucleolus	Upper bound [Min (B, AC)]
T	80.0	385.0	640.0
R	30.0	367.5	580.0
H	50.0	87.5	125.0
F	170.0	250.0	330.0
TR	110.0	752.5	870.0
TH	130.0	472.5	890.0
TF	460.0	635.0	810.0
RH	280.0	455.0	630.0
RF	200.0	617.5	960.0
HF	220.0	337.5	455.0
TRH	360.0	840.0	920.0
TRF	490.0	1,002.5	1,040.0
THF	510.0	722.5	1,060.0
RHF	450.0	705.0	1,010.0
TRHF	1,090.0	1,090.0	1,090.0

an allocation and its bound is reduced or unaffected. For example, the smallest deviation in the benefit-included nucleolus equals \$37,500 as compared with the smallest deviation of \$175,000 in the nucleolus. Second, traditional nucleolus allocations can breach benefit-included upper bounds (but not the lower bounds). For example, the nucleolus allocation of \$237,500 (Table 2) to hunting clearly exceeds the upper bound of \$125,000 defined by its benefit. The benefit-included nucleolus allocations are always consistent with the benefit-included bounds.

Nucleolus and benefit-included nucleolus allocations are not necessarily better or worse than other allocations in the core. Nucleolus and nucleolus modifications are the only currently available means of generating unique core-consistent allocations (Rideout and Hoff 1987).

CONCLUSIONS

Allocations consistent with the logic of game theory define a rational or Pareto-efficient core. This core is equivalent to the full set of separable costs, and hence provides consistency with the early water resources literature based on cost separability. However, the core accounts for the separable costs of groups of purposes as well as individual purposes.

This paper incorporated the early water resources principle that allocations should not exceed benefits into the framework of game theory by defining a benefit-included core. The benefit-included core ensures that no allocation can exceed the lesser of purpose benefit and alternative cost. In the benefit-included core, upper allocation bounds no longer define the separable cost set, but each separable cost is covered. Specific allocations consistent with the benefit-included core can be accomplished using the benefit-included nucleolus.

An application to a multipurpose forest road illuminated the previously obscure programming procedure of the nucleolus. It also illustrated a similar procedure for applying the benefit-included nucleolus. The nucleolus, refined in the context of the benefit-included core, maintains its notable property of ensuring core-consistent allocations.

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